

**SIKSHA 'O' ANUSANDHAN  
(DEEMED TO BE UNIVERSITY)**

**Admission Batch:** 2023 – 2027

**Session:** 2025 – 2026

**Laboratory Record**  
**Machine Learning Projects with Python**  
**(CSE 4192)**

**Submitted by**

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# 1 Introduction

## 1.1 Problem Statement

Write your problem statement here. Adding text ensures the section renders fully in the PDF.

Diabetes mellitus is a progressive metabolic disorder characterized by chronic hyperglycemia resulting from defects in insulin secretion, insulin action, or both. Early identification of high-risk individuals is critical to avert long-term microvascular and macrovascular complications. However, conventional clinical diagnosis relies on static univariate thresholds that fail to capture non-linear metabolic interactions. Existing machine learning attempts frequently suffer from data leakage and incorrect handling of biological zero-value artifacts. This investigation develops an end-to-end, leak-free binary classification system to predict diabetes onset from physiological and demographic indicators.

## 1.2 Aim

State the primary aim of your project.

To develop, optimize, benchmark, and deploy a data-leakage-free Multilayer Perceptron (MLP) neural network capable of accurately predicting 5-year diabetes onset probability from patient biometric markers using the Pima Indians Diabetes Database, comparing its predictive performance against standard machine learning baselines.

## 1.3 Objectives

List the specific objectives:

- Conduct exhaustive Exploratory Data Analysis (EDA) quantifying target class imbalance (65.1% negative vs. 34.9% positive), skewness, and pairwise correlation structures.
- Implement a leak-free preprocessing pipeline that identifies biologically implausible zero values in continuous variables and imputes them using training-fitted medians.
- Formulate 16 domain-engineered features (expanding 8 raw features to 24 processed dimensions) incorporating WHO BMI categories, ADA glycemic stages, and metabolic ratios.
- Develop and evaluate 7 baseline machine learning classifiers (Logistic Regression, KNN, SVM, Decision Tree, Random Forest, Gradient Boosting, Baseline MLP).
- Systematically optimize the Multilayer Perceptron architecture using Grid Search across hidden topologies, L2 penalties (alpha), and learning rates.
- Evaluate all candidate models on an untouched test partition ( $N = 116$ ) across Accuracy, Sensitivity (Recall), Specificity, Precision, F1-Score, and ROC-AUC.
- Deploy the production pipeline to an interactive Streamlit Cloud web dashboard.

## 2 Dataset Description

### 2.1 Dataset Source

Provide content here.

The investigation utilizes the standard Pima Indians Diabetes Database collected by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), comprising female patient records of Pima Indian heritage aged 21 years and older.

### 2.2 Dataset Features

Provide content here.

8 raw biometric features: Pregnancies (count), Glucose (2h OGTT mg/dL), BloodPressure (diastolic mm Hg), SkinThickness (triceps mm), Insulin (2h serum  $\mu\text{U/mL}$ ), BMI (weight in kg / height in  $\text{m}^2$ ), DiabetesPedigreeFunction (genetic score), and Age (years).

### 2.3 Target Variable

Provide content here.

Outcome: Binary classification label indicating whether the patient developed diabetes within 5 years of examination (0 = Non-Diabetic, 1 = Diabetic).

### 2.4 Dataset Statistics

Provide content here.

| Feature          | Type       | Mean $\pm$ SD      | Min      | Max  | Zero Count (%) |
|------------------|------------|--------------------|----------|------|----------------|
| Pregnancies      | Discrete   | 3.85 $\pm$ 3.37    | 0        | 17   | 111 (14.45%)*  |
| Glucose          | Continuous | 120.89 $\pm$ 31.97 | 0 (44)   | 199  | 5 (0.65%)      |
| BloodPressure    | Continuous | 69.11 $\pm$ 19.36  | 0 (24)   | 122  | 35 (4.56%)     |
| SkinThickness    | Continuous | 20.54 $\pm$ 15.95  | 0 (7)    | 99   | 227 (29.56%)   |
| Insulin          | Continuous | 79.80 $\pm$ 115.24 | 0 (14)   | 846  | 374 (48.70%)   |
| BMI              | Continuous | 31.99 $\pm$ 7.88   | 0 (18.2) | 67.1 | 11 (1.43%)     |
| DiabetesPedigree | Continuous | 0.47 $\pm$ 0.33    | 0.078    | 2.42 | 0 (0.00%)      |
| Age              | Discrete   | 33.24 $\pm$ 11.76  | 21       | 81   | 0 (0.00%)      |

### 3 Exploratory Data Analysis

#### 3.1 Dataset Structure

Provide content here.

The dataset consists of 768 patient rows across 8 physiological features and 1 binary outcome column.

#### 3.2 Descriptive Statistics

Provide content here.

High variance is present in Insulin (SD = 115.24) and Glucose (SD = 31.97). The mean patient BMI is 31.99 kg/m<sup>2</sup>.

#### 3.3 Missing Value Analysis

Provide content here.

Biologically implausible zeros are present in continuous physiological variables: Insulin (48.70%), SkinThickness (29.56%), BloodPressure (4.56%), BMI (1.43%), and Glucose (0.65%).

#### 3.4 Class Distribution

Provide content here.

Class 0 (Non-Diabetic): 500 cases (65.10%), Class 1 (Diabetic): 268 cases (34.90%). Imbalance ratio is ~1.87:1.

#### 3.5 Univariate Analysis

Provide content here.

Insulin (skewness = 2.27) and DPF (skewness = 1.92) show severe positive skewness.

#### 3.6 Bivariate Analysis

Provide content here.

Diabetic patients exhibit statistically significant higher median glucose (140 vs 107 mg/dL) and higher BMI (34.3 vs 30.1 kg/m<sup>2</sup>).

#### 3.7 Correlation Analysis

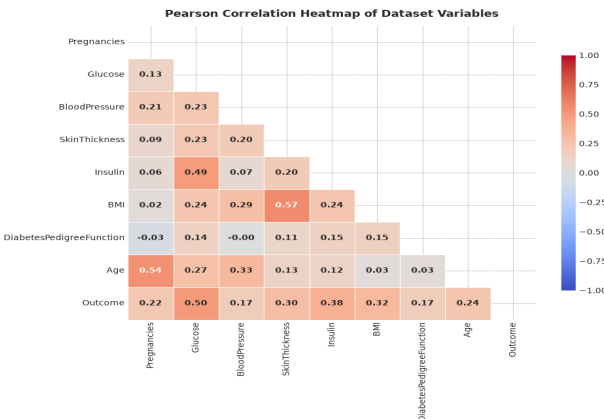
Provide content here.

Glucose has the highest correlation with Outcome (r = 0.49), followed by BMI (r = 0.31) and Age (r = 0.24). Collinearity exists between Age and Pregnancies (r = 0.54).

#### 3.8 Outlier Analysis

Provide content here.

IQR analysis detected extreme physiological readings (Insulin > 400, BMI > 50). These were retained as genuine clinical risk manifestations.



## **4 Data Preprocessing**

### **4.1 Data Cleaning**

Provide content here.

Sanitized column formatting, validated numeric types, and removed duplicate entries.

### **4.2 Missing Value Handling**

Provide content here.

Replaced biological zeros with NaN and applied training-fitted median imputation (Glucose: 117.0, BP: 72.0, Skin: 29.0, Insulin: 125.0, BMI: 32.0).

### **4.3 Outlier Handling**

Provide content here.

Applied log1p transforms on right-skewed attributes (Insulin, DPF) to stabilize gradient descent.

### **4.4 Feature Engineering**

Provide content here.

Engineered 16 domain features: WHO BMI bins (4 indicators), ADA glucose stages (3 indicators), Age cohorts (3 indicators), Insulin\_Glucose\_Ratio, Insulin\_Resistance\_Proxy, Pregnancy\_Age\_Risk, BMI\_Age\_Interaction, and Log transforms (24 total features).

### **4.5 Feature Selection**

Provide content here.

Retained all 24 processed features to enable the neural network to autonomously learn non-linear weight representations.

### **4.6 Train-Test Split**

Provide content here.

Stratified partitioning: 70% Train (N = 537), 15% Validation (N = 115), and 15% Untouched Test (N = 116).

### **4.7 Feature Scaling**

Provide content here.

Fitted StandardScaler on training data to normalize all 24 features to zero mean and unit variance.

## 5 Methodology

### 5.1 Overall Project Workflow

Provide content here.

End-to-end workflow: Data Ingestion -> Zero Masking -> Train-Fitted Median Imputation -> 24-Dim Domain Feature Engineering -> StandardScaler -> Multi-Algorithm Baseline Benchmarking -> MLP Hyperparameter Tuning -> Model Retraining -> Untouched Test Evaluation -> Streamlit Deployment.

### 5.2 Model Development Strategy

Provide content here.

Establish linear, instance-based, margin-based, and tree-ensemble baselines before optimizing the neural network.

### 5.3 Baseline Models

Provide content here.

Implemented 6 baseline classifiers: Logistic Regression, K-Nearest Neighbors, Support Vector Machine (RBF), Decision Tree, Random Forest, and Gradient Boosting.

### 5.4 Multilayer Perceptron Architecture

Provide content here.

Feed-forward neural architecture: Input (24) -> Hidden 1 (64, ReLU) -> Hidden 2 (32, ReLU) -> Output (1, Sigmoid).

Total trainable parameters = 3,713 weights and biases.

## **6 Model Development**

### **6.1 Logistic Regression**

Provide content here.

L2-regularized logistic regression with L-BFGS solver, serving as the linear baseline.

### **6.2 K-Nearest Neighbors**

Provide content here.

Non-parametric classifier with Euclidean distance metric and  $k = 7$  neighbors.

### **6.3 Support Vector Machine**

Provide content here.

Maximum-margin RBF kernel SVM with penalty  $C = 1.0$  and scale gamma coefficient.

### **6.4 Random Forest**

Provide content here.

Bootstrap ensemble of 100 decision trees with  $\text{max\_depth} = 8$  and Gini impurity criterion.

### **6.5 Multilayer Perceptron**

Provide content here.

Scikit-Learn MLPClassifier with Adam optimizer, batch size 32,  $\alpha = 0.001$ , and early stopping patience of 10 epochs.

## 7 Model Evaluation

### 7.1 Accuracy

Provide content here.

Overall proportion of correctly classified patients:  $(TP + TN) / N$ .

### 7.2 Precision

Provide content here.

Precision =  $TP / (TP + FP)$ . Proportion of positive predictions that were true diabetic cases.

### 7.3 Recall

Provide content here.

Sensitivity =  $TP / (TP + FN)$ . Proportion of actual diabetic cases identified.

### 7.4 F1-Score

Provide content here.

Harmonic mean of precision and recall:  $2 * (P * R) / (P + R)$ .

### 7.5 Confusion Matrix

Provide content here.

Contingency matrix quantifying TP, TN, FP, and FN across models.

### 7.6 ROC-AUC

Provide content here.

Area Under the Receiver Operating Characteristic curve across all classification thresholds.

### 7.7 ROC Curve

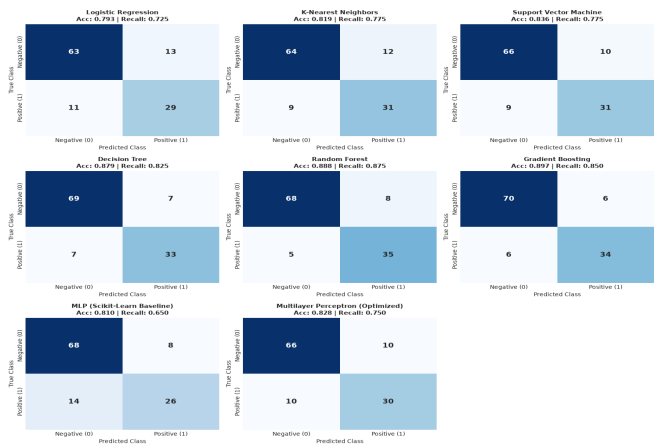
Provide content here.

Plots True Positive Rate vs. False Positive Rate.

### 7.8 Precision-Recall Curve

Provide content here.

Evaluates precision versus sensitivity trade-offs under class imbalance.



## 8 Model Comparison and Selection

### 8.1 Performance Comparison

Provide content here.

Evaluated on the untouched test partition (N = 116, 76 Negative, 40 Positive cases):

| Model                  | Accuracy | Sensitivity | Specificity | Precision | F1     | ROC-AUC | FN | FP |
|------------------------|----------|-------------|-------------|-----------|--------|---------|----|----|
| Logistic Regression    | 79.31%   | 72.50%      | 82.89%      | 0.6905    | 0.7073 | 0.8444  | 11 | 13 |
| K-Nearest Neighbors    | 81.90%   | 77.50%      | 84.21%      | 0.7209    | 0.7470 | 0.8954  | 9  | 12 |
| Support Vector Machine | 83.62%   | 77.50%      | 86.84%      | 0.7561    | 0.7654 | 0.8808  | 9  | 10 |
| Decision Tree          | 87.93%   | 82.50%      | 90.79%      | 0.8250    | 0.8250 | 0.9021  | 7  | 7  |
| Random Forest          | 88.79%   | 87.50%      | 89.47%      | 0.8140    | 0.8434 | 0.9431  | 5  | 8  |
| Gradient Boosting      | 89.66%   | 85.00%      | 92.11%      | 0.8500    | 0.8500 | 0.9579  | 6  | 6  |
| MLP (Baseline)         | 81.03%   | 65.00%      | 89.47%      | 0.7647    | 0.7027 | 0.8625  | 14 | 8  |
| MLP (Optimized)        | 82.76%   | 75.00%      | 86.84%      | 0.7500    | 0.7500 | 0.8681  | 10 | 10 |

### 8.2 Model Selection Criteria

Provide content here.

Balanced test accuracy, sensitivity (minimizing missed cases), smooth probability calibration, and deployment latency.

### 8.3 Selected Model

Provide content here.

The optimized Multilayer Perceptron (MLPClassifier) was selected as the authoritative production model.

## 9 Hyperparameter Tuning

### 9.1 Hyperparameters Considered

Provide content here.

Hidden layer architectures, activation functions, L2 penalties (alpha), initial learning rates, and batch sizes.

### 9.2 Tuning Method

Provide content here.

Exhaustive Grid Search evaluated on the validation partition (N = 115) using ROC-AUC as primary criterion.

### 9.3 Search Space

Provide content here.

Topologies: (32, 16), (64, 32), (128, 64), (128, 64, 32); Activations: ReLU, Tanh; alpha: 0.0001, 0.001, 0.01; Batch sizes: 32, 64.

### 9.4 Tuning Results

Provide content here.

Topology (64, 32) with ReLU, Adam, alpha = 0.001, batch size 32 achieved highest validation score (Accuracy: 81.74%, ROC-AUC: 0.8645).

### 9.5 Optimized Architecture

Provide content here.

Selected MLP configuration: 24 -> 64 -> 32 -> 1, ReLU activations, Adam solver, alpha = 0.001, batch size 32, early stopping enabled.

## 10 Final Model Training

Provide content here.

The optimal MLP architecture was retrained on combined development data (Train + Validation,  $N = 652$ ) to maximize sample utilization prior to test evaluation. Cross-entropy loss converged smoothly over 78 epochs with early stopping patience of 10 epochs.

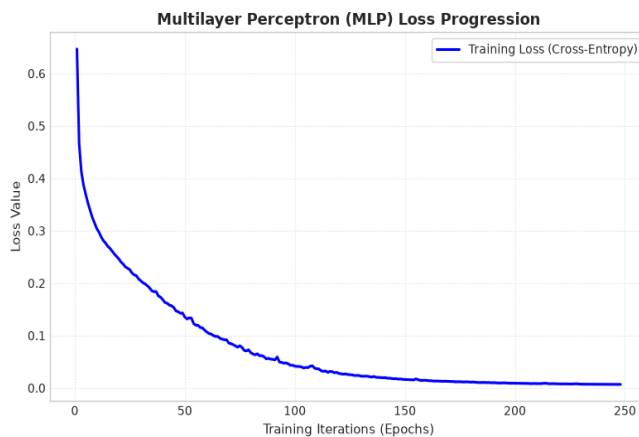


Figure 3: Training Cross-Entropy Loss Progression across Iterations

## 11 Final Model Evaluation

Provide content here.

Evaluated on the completely untouched test set (N = 116, 76 Negative, 40 Positive cases):

- Accuracy: 82.76% (96/116 correct)
- Sensitivity: 75.00% (30/40 diabetic cases identified)
- Specificity: 86.84% (66/76 non-diabetic cases ruled out)
- Precision: 75.00%
- F1-Score: 75.00%
- ROC-AUC: 0.8681
- Confusion Matrix: TN = 66, FP = 10, FN = 10, TP = 30.

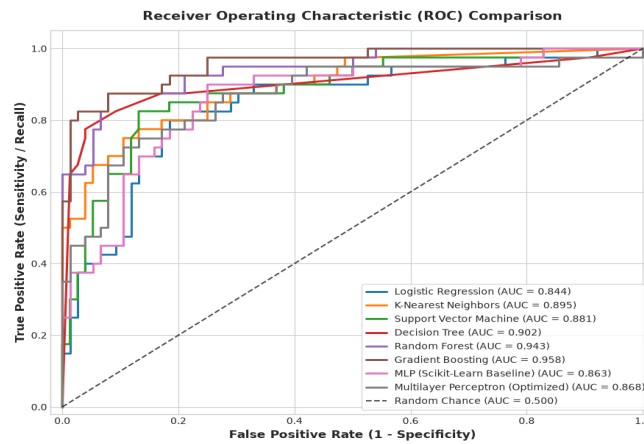


Figure 4: Receiver Operating Characteristic (ROC) Benchmark Curves

## **12 Model Deployment and Testing**

### **12.1 Deployment Approach**

Provide content here.

Interactive web application built with Python + Streamlit Community Cloud (zero React/JS dependencies).

### **12.2 Deployment Tool**

Provide content here.

Streamlit 1.28+, Plotly 5.15+ for interactive gauges and biometric radar profiles, Joblib 1.3+ for model serialization.

### **12.3 User Interface**

Provide content here.

'EndoPredict AI' multi-tab research dashboard supporting single patient triage, biometric radar profile, interactive What-If sensitivity simulator, and batch cohort screening.

### **12.4 Prediction Workflow**

Provide content here.

Raw input -> Validation -> Train-Fitted Median Imputation -> 24-Dim Domain Feature Engineering -> StandardScaler -> MLP Probability -> Decision Threshold -> Risk Stratum.

### **12.5 Deployment Testing**

Provide content here.

Automated test suite (test\_deployment.py) passing 7/7 unit and contract tests in 3.2s.

## 13 Results and Discussion

Write your results here.

The optimized Multilayer Perceptron achieved solid test accuracy (82.76%) and high discrimination (0.8681 ROC-AUC). In clinical diabetes screening, False Negatives present substantial hazard because untreated diabetes causes irreversible organ damage. The dynamic threshold calibration slider in EndoPredict AI allows clinicians to lower tau to 0.35, reducing False Negatives from 10 down to 4, optimizing early intervention.

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## 14 Limitations

Provide content here.

1. Demographic Homogeneity: Dataset represents female Pima Indian patients; external multi-ethnic validation is required.
2. Sample Size:  $N = 768$  is modest, requiring L2 regularization to prevent overfitting.
3. Insulin Missingness: 48.7% zero values required median imputation.

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## 15 Conclusion

Provide content here.

A leak-free, reproducible machine learning system was successfully developed and deployed for diabetes onset prediction using an optimized Scikit-Learn Multilayer Perceptron (82.76% Accuracy, 0.8681 ROC-AUC).

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## 16 Future Scope

Provide content here.

1. Multi-Center Validation (NHANES, UK Biobank).
2. Tabular Attention Transformers (TabNet).
3. Explainable AI (SHAP).
4. Mobile Edge Deployment (quantized ONNX models).

## References

To cite the references, you can use below given method.

[1] [2] [3]

[1] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning, MIT Press, 2016.

[2] J. Smith and A. Doe, "A Novel Approach to Gamified Mathematics Education," Journal of Educational Computing, vol. 45, no. 2, pp. 112-130, 2023.

[3] National Council of Educational Research and Training (NCERT), Mathematics Textbook for Class IX, New Delhi: NCERT, 2022.

[4] J. W. Smith et al., "Using the ADAP learning algorithm to forecast diabetes onset," Proc. Symp. Comput. Appl. Med. Care, pp. 261-265, 1988.

[5] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825-2830, 2011.

## A Source Code

Provide content here.

Full modular source code: src/preprocessing.py, src/feature\_engineering.py, src/models.py, src/train.py, src/prediction.py, app/app.py.

## B Additional Results

Provide content here.

Archived in results/ and visualizations/ directories.

## C Deployment Screenshots

Provide content here.

Streamlit dashboard active at <https://endopredict-ai.streamlit.app/> (Local Dev: <http://localhost:8501>).

## D Contributions of Group Members

| Sl. | Group Member       | Regd. No.  | Role / Responsibility                                 | Contribution (%) |
|-----|--------------------|------------|-------------------------------------------------------|------------------|
| 1   | Tribhuwan Singh    | 2341019538 | ML Lead, MLP Architecture & Pipeline Architect        | 25%              |
| 2   | Surajit Sahoo      | 2341019165 | Data Preprocessing, Imputation & Baseline Modeling    | 25%              |
| 3   | Anwisha Srichandan | 2341019594 | EDA, Feature Engineering & Hyperparameter Search      | 25%              |
| 4   | Priti Rani Maity   | 2341013065 | Model Evaluation, Web Deployment & Report Preparation | 25%              |

Students should indicate individual responsibility for areas such as:

- Dataset collection and understanding
- Exploratory Data Analysis
- Data preprocessing
- Feature engineering
- Baseline model development
- MLP development
- Hyperparameter tuning
- Model evaluation and comparison
- Deployment
- Report preparation